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The Vanguard of American Magnetics: An Exhaustive Analysis of the Post-2014 Domestic Startup Ecosystem

Executive Summary and Macroeconomic Context

The global supply chain for permanent magnets—critical components embedded in everything from electric vehicle traction motors and artificial intelligence data center cooling systems to F-35 fighter jets and advanced robotics—has long been defined by a near-total monopolization by the People's Republic of China. Historically, China has controlled over 90 percent of the world's rare earth refining and magnet manufacturing capacity, establishing a structural bottleneck that poses severe geopolitical and macroeconomic risks to the United States.¹ For decades, the United States lacked a meaningful domestic magnet manufacturing base. The historical trajectory of the American magnet sector was defined by offshoring and foreign acquisition, highlighted by the 1995 acquisition of General Motors' legacy magnet division, Magnequench, by Chinese-backed entities, and the subsequent transfer of its manufacturing lines overseas by the early 2000s.⁴ Consequently, older legacy companies operating in the American ecosystem, such as Advanced Magnet Lab (founded in 1995), represent a prior era of magnetic research and engineering that predates the current industrial policy imperative.⁶ However, a radical paradigm shift is currently underway. Driven by the urgent imperatives of national security, clean energy transitions, and supply chain resilience, a distinct "new wave" of American magnet manufacturers has emerged. To strictly isolate the newly established companies—specifically startups, joint ventures, and new independent subsidiaries—this exhaustive analysis focuses exclusively on entities founded or launched between 2014 and 2026. This post-2014 cohort represents a definitive break from legacy models, operating strictly as newer market entrants engineered to resurrect the U.S. magnet sector. These new entities are not monolithic in their approach. The data indicates a strategic bifurcation in how these startups are attempting to solve the domestic magnet deficit. The approaches can be categorized into three distinct technological and operational pathways: the material science disruption of engineering rare-earth-free magnets, the development of circular supply chains and advanced recycling, and the heavily capitalized vertical integration of domestic Neodymium Iron Boron (NdFeB) production.

The Policy Architecture and Capital Catalysts

The proliferation of these new companies cannot be understood in a vacuum. Their rapid emergence is the direct result of a synthesized industrial policy architecture that has systematically de-risked capital-intensive hardware manufacturing. Startups in this sector require hundreds of millions of dollars in upfront capital expenditures before achieving

commercial viability—a threshold known in venture capital as the "Valley of Death".⁸

To bridge this formidable gap, the United States government has leveraged a suite of financial and legislative instruments. Key among these are the Defense Production Act (DPA) Title III, the Department of Energy's (DOE) Critical Materials Innovation programs, and the Department of Defense's (DoD) Office of Strategic Capital (OSC).⁹ Crucially, the 2021 National Defense Authorization Act instituted a Defense Federal Acquisition Regulation Supplement (DFARS) requirement stipulating that by 2027, United States weapons systems cannot contain magnets mined, refined, or manufactured in China, Russia, North Korea, or Iran.¹³

This looming January 2027 deadline has created an artificial, guaranteed demand floor.

Defense prime contractors are now legally obligated to source domestic components, prompting venture capital, private equity, and institutional investors—such as Altimeter Capital, One Investment Management (OneIM), Ara Partners, and 1789 Capital—to deploy billions of dollars into these young companies with the assurance of a captive end-market.⁹

Federal Policy / Financial Instrument	Executing Agency	Primary Function in the Magnet Ecosystem	Representative Startup Beneficiary
DFARS 2027 Procurement Ban	Department of Defense	Prohibits Chinese/Russian rare earth magnets in defense platforms, creating guaranteed domestic demand.	Vulcan Elements, USA Rare Earth
Office of Strategic Capital (OSC) Loans	Department of Defense	Provides massive debt financing (up to \$100B authorized) to scale critical hardware infrastructure.	Vulcan Elements, ReElement Tech
DPA Title III Grants	Department of Defense	Direct capital injections to operationalize technical infrastructure and engineer domestic production lines.	e-VAC Magnetics

Section 48c Tax Credits	Department of Treasury / Energy	Qualifying Advanced Energy Project Tax Credits to accelerate domestic clean energy manufacturing facilities.	e-VAC Magnetics
Critical Materials Innovation Funding	Department of Energy	Early-stage grants for material science research, reactor systems, and pilot-scale separation projects.	Niron Magnetics, FeNix Magnetics

Pathway 1: Material Science Disruption and Rare-Earth-Free Alternatives

The most technologically radical approach to addressing the magnet supply chain vulnerability is to engineer the vulnerability out of existence entirely. Rather than competing directly with China on the extraction and refinement of rare earth elements such as neodymium, praseodymium, dysprosium, and terbium, a highly specialized subset of the new startup ecosystem is synthesizing powerful permanent magnets from globally abundant materials.

Niron Magnetics

Founded in 2014 and spun out of foundational research conducted by Distinguished McKnight University Professor Jian-Ping Wang at the University of Minnesota, Niron Magnetics represents the absolute vanguard of rare-earth-free permanent magnet technology.¹⁸

Headquartered in Minneapolis, Minnesota, Niron has commercialized an entirely new class of permanent magnet known commercially as the "Clean Earth Magnet." This technology is based

on the synthesis of Iron Nitride, specifically the alpha double prime phase (α'' - $Fe_{16}N_2$).²⁰

The scientific and macroeconomic implications of this development are profound. Iron and nitrogen are two of the most abundant elements on Earth; industrial scrap iron is ubiquitous, and nitrogen constitutes the primary component of the Earth's atmosphere.³ By relying exclusively on these feedstocks, Niron fundamentally detaches its supply chain from the geopolitical volatility and extreme environmental degradation associated with rare earth mining. The traditional production of one kilogram of neodymium can generate up to 1,000 kilograms of toxic and radioactive waste, severely damaging local ecosystems.²² In stark

contrast, the only byproduct of Niron's manufacturing process is salt water.³

Iron Nitride possesses an exceptionally high theoretical magnetization, offering the potential to match or eventually exceed the performance characteristics of traditional NdFeB magnets across a variety of applications.²¹ Niron's materials are specifically engineered for high-duty-cycle thermal environments. As the rapid adoption of artificial intelligence drives higher-density data center deployments, the demand for highly efficient cooling and thermal management systems has skyrocketed. Niron's magnets offer immense thermal stability, making them ideal for next-generation cooling architectures, server rack cooling, HVAC chillers, and coolant pump motors in AI data centers, as well as electric vehicle traction motors and defense actuators.²¹

To date, Niron has secured over \$100 million in private funding, backed by strategic investments from both venture capital and corporate partners.¹⁸ This private capital was significantly bolstered by a \$17.5 million grant from the United States Department of Energy to specifically scale up its proprietary manufacturing process.¹⁰ The company's roadmap demonstrates a disciplined, highly structured, phased approach to domestic manufacturing scaling.²¹

The first phase consists of a Commercial Pilot Plant currently fully operational in Minneapolis, Minnesota, which is actively manufacturing and sampling products for customers while serving as the critical bridge between research and development and full-scale automation.²¹ The second phase involves a Full-Scale Manufacturing Plant in Sartell, Minnesota, where ground has already been broken. This facility is designed to deliver a manufacturing capacity of 1,500 metric tons per year and is expected to add over 175 full-time manufacturing jobs to the region.²¹ The ultimate third phase is a High-Volume Manufacturing Plant, for which site selection is currently underway. This massive future facility is planned to scale high-volume production up to a capacity of 10,000 metric tons per year, which would solidify Niron as a dominant, indispensable force in domestic industrial infrastructure.²¹

FeNix Magnetics: The Cautionary Tale of the Valley of Death

While Niron Magnetics illustrates the successful trajectory of material science startups, the sector is fraught with severe technical and financial risks. FeNix Magnetics, a startup formerly headquartered in Westlake, Ohio, serves as a vital counterpoint and a demonstration of the extreme difficulty of scaling advanced manufacturing.²⁶

FeNix Magnetics was similarly engaged in the pursuit of rare-earth-free magnets, specifically focusing on the advanced manufacturing of Alpha Double Prime Iron Nitride (ADPIN).²⁴ Led by Chief Technology Officer David Matthiesen, the company sought to develop a fluidized bed reactor system to overcome the rate-limiting steps of nitrided powder production, aiming to synthesize 500-gram sample sizes for subsequent magnet fabrication.²⁴

The startup secured highly competitive federal funding, including backing from the Department of Energy's Advanced Manufacturing Office and the Ohio Federal Research Network, embedding itself in a consortium that included Case Western Reserve University and various aerospace primes.²⁸ The stated goal was to challenge the price-performance

dominance of NdFeB magnets in clean energy applications, specifically targeting a reduction in the massive 90,000 metric ton market share held by traditional rare earth magnets.²⁴ Despite securing federal research grants and producing final technical reports through 2021, the venture ultimately failed to cross the commercialization chasm. The company is currently listed as out of business, its venture capital backing exhausted.²⁶ The dissolution of FeNix Magnetics highlights a fundamental third-order insight: federal grant funding for pilot-scale research is insufficient to guarantee commercial survival. The transition from laboratory synthesis to industrial-scale reactor production involves exponential increases in capital expenditure and engineering complexity, reinforcing why massive, non-recourse debt financing and guaranteed offtake agreements are mandatory for survival in this ecosystem.

Pathway 2: Circular Supply Chains and Advanced Urban Mining

A second cohort of post-2014 startups focuses on an entirely different supply chain vulnerability: the end-of-life cycle of existing magnets. Millions of tons of rare earth magnets currently exist in discarded hard disk drives, decommissioned wind turbines, obsolete MRI machines, and end-of-life electric vehicles across North America and Europe. Pioneering startups are harvesting this "urban mine," creating a domestic, closed-loop circular economy that circumvents traditional upstream mining bottlenecks.

Noveon Magnetics

Founded in 2014 under the name Urban Mining Company before undergoing a comprehensive rebranding in May 2022, Noveon Magnetics is currently the only fully operational manufacturer of sintered NdFeB magnets in the United States.³¹ Headquartered in San Marcos, Texas, the company was established by a uniquely international coalition of engineers and entrepreneurs, including Scott Dunn, Peter Afiuny, Miha Zakotnik (a material scientist from Slovenia), and Catalina Oana Tudor (a computer scientist from Romania).¹⁷

Noveon's primary technological moat is its patented "Magnet-to-Magnet" recycling process. In traditional recycling, the complex magnetic properties and highly specific crystal structures of rare earth alloys are often degraded, heavily oxidized, or lost entirely when the material is shredded and melted down.¹⁷ Noveon's proprietary process, however, preserves and reconstitutes the material without performance degradation, resulting in their branded "EcoFlux" line of high-efficiency permanent magnets.³¹

The EcoFlux production process is vastly superior to traditional rare earth ore mining in terms of its ecological and energetic footprint. It bypasses the toxic separation phases of upstream mining, thereby reducing overall energy consumption by up to 90 percent and slashing the overall environmental footprint by 50 percent.³¹ This powerful combination of uncompromising magnetic performance and deep sustainability has allowed Noveon to lock in highly lucrative, multi-year supply agreements with blue-chip global manufacturers. Most notably, Noveon initiated a massive supply agreement with General Motors in 2025, with deliveries for automotive components commencing in July 2025, alongside major volume agreements with Japanese motor giant Nidec and material sourcing deals with Solvay.¹⁷

Financially, Noveon's trajectory highlights the explosive, compounding growth potential of the circular magnet sector. Operating out of a state-of-the-art 145,000-square-foot facility in San Marcos, the company grew its workforce from a mere 25 employees in 2020 to nearly 100 today.³¹ After generating approximately \$10 million in revenue in 2023—a period when the company was still refining its commercial validation testing—Noveon projects its revenues will surge to an astonishing \$250 million by 2025, representing a compound annual growth rate exceeding 190 percent.³¹ At full operational scale, the company anticipates EBITDA margins of 40 percent at current market pricing.³¹

To finance this hyper-growth, Noveon aggressively courted private equity and venture capital. After early seed funding from familial networks and Small Business Innovation Research (SBIR) grants, the company raised a \$75 million Series B in May 2023, led by NGP Energy Capital Management and Aventurine Partners, valuing the company at approximately \$300 million.¹⁷ This was followed by a massive, milestone \$215 million Series C round in January 2026. This round was anchored by a \$200 million commitment from One Investment Management (OneIM), a global alternative investment firm led by former SoftBank Vision Fund executive Rajeev Misra.¹⁵ The Series C funding is strictly earmarked to scale the San Marcos facility's output to over 2,000 metric tons annually, while also introducing stringent new corporate governance, with OneIM appointing two representatives to Noveon's Board of Directors.³¹

Cyclic Materials

While Noveon Magnetics focuses on manufacturing finished, sintered recycled magnets, Cyclic Materials specializes in a preceding step in the supply chain: the chemical separation and recovery of the raw rare earth elements themselves. Founded in 2021 and originally launching its commercial demo facilities in Kingston, Ontario, the startup is aggressively expanding its footprint into the United States to capture the surging domestic demand for critical minerals.³⁷ Cyclic Materials utilizes advanced, proprietary hydrometallurgical technologies—specifically branded as the MagCycleSM and REEPureSM processes—to extract and recover Mixed Rare Earth Oxides (MREO) from highly complex end-of-life products, such as electric vehicle traction motors and data center electronic waste.³⁷ Recognizing the strategic, policy-driven shift in the United States market, Cyclic announced in January 2026 a massive \$82 million capital investment to establish its largest "Hub" facility to date in McBee, South Carolina.³⁹ This South Carolina facility will initially possess the capacity to process 2,000 metric tons of raw magnet material to produce 600 tonnes of MREO annually. Furthermore, the company has explicitly planned infrastructure expansions to push this capacity to 6,000 tonnes of processing, yielding 1,800 tonnes of MREO per year.³⁹ To contextualize the scale of this output, 1,800 tonnes of MREO supplies enough rare earth material to manufacture the equivalent of 6 million hybrid vehicle transmissions annually.³⁹ Cyclic Materials has also demonstrated a high degree of sophisticated supply chain synergy, embedding itself deeply within the new domestic ecosystem. In late 2025, the company signed a highly strategic 10-year exclusive agreement with e-VAC Magnetics—another newly established U.S. magnet manufacturer operating in the same state.³⁹ Under this agreement,

Cyclic Materials will recycle 100 percent of the manufacturing byproducts, colloquially known as "swarf," generated at e-VAC's new factory in Sumter, South Carolina.³⁹

This partnership highlights a profound third-order insight into the U.S. re-shoring strategy: the intentional creation of highly localized, highly efficient industrial ecosystems. By linking their operations, the waste stream of one domestic facility becomes the vital feedstock for another. This drastically reduces logistical overhead, eliminates the cost of hazardous waste disposal for e-VAC, provides a guaranteed supply of high-grade inputs for Cyclic, and effectively insulates both startups from external, trans-Pacific supply chain shocks.

Startup	Founding Year	Technological Approach	Key Facilities & Hubs	Core Product / Output
Noveon Magnetics	2014	Magnet-to-Magnet Recycling	San Marcos, TX (145k sq ft)	EcoFlux Sintered NdFeB
Cyclic Materials	2021	Hydrometallurgical Separation	McBee, SC & Kingston, ON	Mixed Rare Earth Oxides
Niron Magnetics	2014	Material Science Substitution	Minneapolis & Sartell, MN	Iron Nitride (α'' - $Fe_{16}N_2$)

Pathway 3: The Vertically Integrated Sintered NdFeB Titans

The third, most capital-intensive pathway involves establishing traditional, ground-up manufacturing capacity for sintered NdFeB magnets on American soil. Sintered NdFeB magnets remain the most powerful permanent magnets commercially available and are deemed absolutely critical for modern defense systems, aerospace applications, and advanced robotics.¹⁴

However, manufacturing them is exceptionally difficult. The process requires powdering rare earth alloys to 3-5 microns in size (roughly 20 times finer than a human hair) in strict oxygen-restricted environments, wet-pressing the powder into blocks within a magnetic field to align the crystal structures, and then sintering (baking) the blocks at highly precise temperatures before machining and coating.⁴¹ Because of these immense complexities in metallurgy and thermal dynamics, startups in this category require astronomical capital expenditures. Consequently, they often pursue vertical integration—attempting to control the

entire process from the raw upstream mineral refinement down to the final downstream magnet fabrication.

Vulcan Elements

Incorporated in 2023 and headquartered in the Research Triangle Park region of North Carolina—with operational footprints spanning Durham, Raleigh, and Benson—Vulcan Elements has emerged as one of the fastest-scaling industrial hardware startups in United States history.⁹ Founded by CEO John Maslin, a former U.S. Navy Nuclear Propulsion Program officer who previously managed the complex finances for the Ballistic Missile Submarine Program, the company is hyper-focused on securing the defense industrial base and resolving national security vulnerabilities.⁴²

Vulcan's rise has been meteoric, driven by unprecedented government backing and elite private equity. In March 2025, the company officially opened a 21,000-square-foot commercial and research and development facility in Raleigh, directly supporting the tactical onshoring of traceable NdFeB magnets.⁴² The launch was attended by influential thought leaders, including Dr. Chris Miller, author of *Chip War*, underscoring the strategic parity between semiconductor manufacturing and rare earth magnetism.⁴³

By late 2025 and early 2026, the company achieved financial milestones that fundamentally redefined the capitalization limits of the sector:

1. **Venture Capital Acceleration:** Vulcan raised a \$65 million Series A round led by Altimeter Capital—a premier technology investment firm with over \$12 billion in assets under management, steered by Brad Gerstner. The round saw significant participation from OnellM, the same mega-fund backing Noveon.¹³
2. **Unprecedented DoD Financing:** In November 2025, Vulcan secured a staggering \$620 million loan from the Pentagon's Office of Strategic Capital (OSC). This represented the largest loan ever issued by the OSC, authorized under the "One Big Beautiful Bill Act," a federal legislative package providing up to \$100 billion in lending to accelerate domestic critical mineral production.⁹
3. **The Mega-Factory Expansion:** Leveraging this massive capital influx, Vulcan announced a \$1.4 billion comprehensive partnership with the U.S. government to scale operations. The company selected Benson, North Carolina, for a \$1 billion, 1-million-square-foot manufacturing facility. This gargantuan plant is expected to create 1,000 high-paying manufacturing jobs.⁴⁶

The ultimate production target for Vulcan Elements is an astounding 10,000 metric tons of NdFeB magnets annually by the end of the decade, a scale designed to single-handedly close the defense supply gap.⁹ However, the unprecedented speed of Vulcan's capitalization highlights the increasingly blurred lines between national security policy and private venture capital, which occasionally introduces political risk. In August 2025, 1789 Capital, a firm heavily backed by Donald Trump Jr., invested significant equity into Vulcan just three months before the second Trump administration's Pentagon awarded the company its record-breaking \$620 million loan.⁹ While this raised public questions regarding conflicts of interest, Vulcan's aggressive expansion remains the absolute cornerstone of the government's strategy to meet

the strict 2027 DFARS procurement deadline.⁹

USA Rare Earth (USAR)

Founded in 2019 by mining executive Pini Althaus, USA Rare Earth embodies the absolute realization of the "mine-to-magnet" philosophy.⁴⁹ Now headquartered in Stillwater, Oklahoma, and currently led by CEO Barbara Humpton, the company is relentlessly building a fully vertically integrated supply chain, aggressively insulated from foreign interference.¹⁴

Unlike startups that rely solely on recycled feedstocks or third-party international suppliers, USAR controls the foundational upstream mining rights to the Round Top heavy rare earth and critical minerals deposit in West Texas. This deposit is uniquely rich in 15 of the 17 rare earth elements, as well as critical tech-metals like lithium and gallium.¹⁴ To secure the highly complex midstream processing capabilities, the company embarked on a rapid, opportunistic acquisition spree. Most notably, USAR acquired the UK-based metal and alloy production company Less Common Metals (LCM) in late 2015, effectively importing decades of European metallurgical expertise, and subsequently took a 12.5 percent equity stake in the French rare earths processor Carester.¹⁴

The culmination of this relentless vertical integration is currently taking place at USAR's massive 310,000-square-foot downstream manufacturing facility in Stillwater, Oklahoma. The company achieved a major technical milestone in January 2025 by successfully producing its first batch of commercial-grade sintered permanent rare earth magnets at its on-site Innovations Lab.¹⁴

Operating with rapid momentum, by the first quarter of 2026, USAR began the complex commissioning of Production Line 1a, which possesses a 1,200-metric-ton annual capacity. The long-term strategic plan is to scale this infrastructure across four distinct production lines, aiming for a maximum, full-scale capacity of nearly 5,000 metric tons annually.¹⁴

To fund this highly capital-intensive model, USAR has operated aggressively within both private and public financial markets. The company, which achieved a \$1 billion valuation within its first five years under Althaus, secured a monumental \$1.6 billion in conditional funding from the U.S. government to assist in financing both the Texas mine and the Oklahoma factory.¹⁴

Furthermore, to access broader pools of institutional capital, the company initiated structural processes to become a publicly traded entity via a Special Purpose Acquisition Company (SPAC) merger with Inflection Point Acquisition Corp. II, targeting a total addressable market estimated at \$41.1 billion by 2035.⁵²

e-VAC Magnetics

Formed in 2023 as a wholly owned U.S. subsidiary of the global private equity firm Ara Partners and its portfolio company Vacuumschmelze (VAC), e-VAC Magnetics serves as a prime example of premium foreign technical expertise being strategically localized, capitalized, and deployed as a newly minted American startup.¹⁶

e-VAC is actively constructing a state-of-the-art permanent magnet manufacturing facility in Sumter County, South Carolina. Breaking ground in March 2024 and officially opening operations in the fall of 2025, the facility is actively scaling to full industrial capacity throughout

2026 to produce mission-critical NdFeB magnets.⁵⁵ The formation and immediate success of e-VAC were heavily catalyzed by two foundational agreements that provided both commercial and governmental certainty:

1. **The Commercial Anchor:** In January 2023, e-VAC signed a binding, 10-year supply agreement with General Motors (GM). Under this contract, e-VAC will manufacture permanent magnets, utilizing locally sourced raw materials, for the electric motors utilized across a broad portfolio of GM's North American electric vehicles.¹⁶
2. **The Defense Anchor:** In September 2023, the Department of Defense awarded e-VAC \$94.1 million under the DPA Title III program (with \$31.4 million in non-federal cost-matching). This massive grant was explicitly designed to allow e-VAC to operationalize technical infrastructure, acquire specialized manufacturing equipment, and engineer high-volume production lines capable of supplying components for platforms like the F-35.¹¹

The financial architecture constructed behind e-VAC is formidable and highly sophisticated. Ara Partners successfully secured \$335 million in non-recourse financing in 2024—a landmark deal led by BMO and Mitsubishi UFJ Financial Group, with participation from CIBC, Rabobank, and Société Générale.¹⁶ This complex structuring was heavily praised by financial institutions, winning both the IJ Global "2024 Energy Transition Deal of the Year" and Proximo's "North American Manufacturing Deal of the Year".⁵⁸ Additionally, e-VAC was awarded a highly competitive \$111.9 million Qualifying Advanced Energy Project Tax Credit (Section 48c), perfectly demonstrating how targeted federal tax incentives are directly materializing as ground-up, heavy manufacturing capabilities.⁵⁵

MP Materials (Magnetics Division and 10X Campus)

While MP Materials is globally recognized as the operator of the legacy Mountain Pass rare earth mine in California, its downstream *Magnetics Division* operates functionally as a brand-new, high-growth hardware startup within the broader corporate structure. For decades, MP Materials was structurally forced to ship its mined rare earth oxides across the Pacific to China for complex refining and metallization. To definitively break this cycle, the company recently established a massive new 250,000-square-foot downstream facility named "Independence" in Fort Worth, Texas.⁶⁰

The Independence facility is a testament to rapid industrial scaling. It began commercial metal production in 2024 and quickly advanced to finished sintered NdFeB magnet production in 2025, operating on commercial equipment with the intent to scale toward a pilot capacity of 1,000 metric tons per year.⁶⁰ However, the true "new company" milestone occurred in February 2026, when MP Materials announced the spectacular development of "10X"—a sprawling new 120-acre U.S. rare earth magnet manufacturing campus located in Northlake, Texas.⁶⁴

The 10X campus represents a staggering \$1.25 billion capital investment designed to catapult MP's production capacity to 10,000 metric tons of NdFeB magnets annually.⁶⁴ Backed by significant DoD partnerships—which provide the essential long-term demand certainty required for such a massive build-out—and an impressive \$500 million offtake agreement with major commercial partners, 10X will create over 1,500 direct manufacturing and engineering

jobs.⁶²

The State of Texas heavily incentivized this rapid expansion, providing a highly attractive economic package that included a \$12.8 million grant from the Texas Enterprise Fund and a massive \$53.4 million grant from the Texas Semiconductor Innovation Fund.⁶⁴ Engineering and equipment procurement for the 10X campus is well underway, with full commissioning slated for 2028, aligning perfectly with the DoD's mandate to definitively purge Chinese components from the defense industrial base.⁶³

ReElement Technologies and the POSCO Joint Venture

ReElement Technologies, operating as a nimble subsidiary of the NASDAQ-listed American Resources Corporation, represents a highly critical piece of the midstream refining puzzle. The company focuses exclusively on the chromatographic purification and separation of critical minerals, uniquely processing both battery black mass (LFP and NMC chemistries) and end-of-life rare earth feedstocks.⁶⁶

In May 2026, ReElement firmly cemented its status in the new magnet ecosystem by signing a landmark \$200 million agreement with the massive South Korean industrial conglomerate POSCO International to establish a dedicated joint venture within the United States.¹ Dubbed the "Boiler Maker Project," this joint venture aims to build the very first fully integrated rare earth separation, refining, and permanent magnet production complex in the U.S..⁶⁸

With POSCO International serving as the majority shareholder and guiding the commercial management, and ReElement providing the proprietary core separation and refining technology, the JV plans to build a facility with an initial capacity of 6,000 tons of rare earth separation.¹ The \$200 million total investment is structured to be deployed in two distinct phases: utilizing \$100 million for initial plant construction, equipment procurement, and early-stage operations, and reserving the remaining \$100 million for subsequent aggressive expansions into the direct, high-volume manufacturing of permanent magnets.¹

Bolstered by an additional \$80 million capital allocation directed to ReElement via the DoD's Office of Strategic Capital, this joint venture demonstrates a profound shift in global strategy: the vital role of international alliances—specifically deep U.S.-Korea industrial cooperation—in expediting the domestic supply chain restructuring and insulating allied nations from Chinese export controls.⁴⁵

Startup / Joint Venture	Foundational Year	Sintered NdFeB Target Capacity	Key Capitalization / Financing Mechanisms	Location Hub
Vulcan Elements	2023	10,000 metric tons/year	\$620M OSC Loan, \$65M Series A	Benson, NC

MP Materials (10X)	2026	10,000 metric tons/year	\$1.25B Investment, Texas State Grants	Northlake, TX
USA Rare Earth	2019	5,000 metric tons/year	\$1.6B Govt Support, SPAC Merger	Stillwater, OK
ReElement / POSCO JV	2026	Integrated (6,000 tons refining)	\$200M JV Investment, \$80M OSC Loan	United States
e-VAC Magnetics	2023	Commercial Scale (Classified)	\$335M Financing, \$111M Tax Credit	Sumter, SC

Niche Applications: Contactless Magnetics in Aerospace

Beyond the massive, high-volume industrial scaling of sintered NdFeB and Iron Nitride magnets, the post-2014 startup wave also includes highly specialized, agile niche players targeting specific high-tech hardware bottlenecks.

FluxWorks

Founded in 2021 by CEO Bryton Praslicka initially in College Station, Texas, and subsequently relocated to Conroe, Texas, in 2024, FluxWorks is commercializing highly advanced magnetic gears.⁷¹ The company specializes in contactless magnetic gears designed specifically for extreme environments, with a primary focus on space applications and in-space manufacturing. Traditional mechanical gears require physical contact and continuous lubricants, both of which are highly problematic and prone to rapid failure in the microgravity and extreme temperature fluctuations of outer space.⁷¹

FluxWorks' flagship "Celestial Gear" technology relies entirely on complex magnetic interactions, completely eliminating physical wear, debris generation, and the need for fluid lubrication. In 2026, the company successfully closed a \$5 million seed round led by Scout Ventures and the Michigan Capital Network.⁷¹ This funding was facilitated in part by the Department of Defense and the Small Business Administration's Small Business Investment

Company (SBIC) Critical Technologies Initiative.⁷¹

FluxWorks illustrates a vital component of the domestic magnet renaissance: it is not merely about brute-force replication of foreign high-volume manufacturing at scale, but also about pioneering entirely novel, high-margin applications designed specifically to maintain American supremacy in aerospace and national security.

Thematic Analysis: The Emergence of Geographic Innovation Hubs

A macro-level analysis of the geographical distribution of these new startups reveals a highly structured, deliberate clustering effect. Rather than being distributed evenly or randomly across the United States, the new magnet industry is coagulating into distinct regional industrial hubs. This clustering is not coincidental; it allows for the sharing of highly specialized labor, proximity to complementary hardware industries, and the creation of highly localized, closed-loop supply chains.

The Texas Rare Earth Triangle

Texas is rapidly establishing itself as the absolute epicenter of U.S. rare earth and magnet manufacturing, supported heavily by aggressive, business-friendly state-level incentive funds.

- **San Marcos:** Serves as the home to Noveon Magnetics and its highly advanced 145,000-square-foot recycling and manufacturing facility.³¹
- **Fort Worth / Northlake:** Functions as the heavy manufacturing core, hosting MP Materials' Independence facility and the upcoming massive \$1.25 billion 10X manufacturing campus.⁶⁵
- **West Texas:** The location of USA Rare Earth's Round Top mining operation, providing a localized source of raw heavy rare earths.⁵¹
- **Conroe:** The headquarters of FluxWorks, integrating the aerospace application layer.⁷¹

This dense concentration of downstream manufacturing, midstream refining, and upstream mining allows Texas to operate as a nearly self-sufficient industrial ecosystem, highly resistant to global logistical disruptions.

The Carolinas Magnet Corridor

A secondary, equally potent hub has emerged across North and South Carolina, driven by proximity to deep-water ports, affordable industrial land, and strong, non-unionized manufacturing labor pools.

- **North Carolina (Durham/Raleigh/Benson):** Vulcan Elements has established its corporate headquarters, its high-tech R&D center, and its planned \$1 billion, 10,000-ton-per-year mega-facility in the Research Triangle and surrounding strategic counties.⁹
- **South Carolina (Sumter/McBee):** e-VAC Magnetics is rapidly scaling its massive NdFeB facility in Sumter County.⁵⁶ Just a short logistical distance away in McBee, Cyclic Materials is executing its \$82 million investment into a rare earth recycling Hub.³⁹

The Carolinas Corridor exemplifies the third-order macroeconomic benefits of industrial agglomeration. Cyclic Materials' strategic decision to locate its massive recycling Hub in South Carolina was directly influenced by e-VAC's presence. By signing a 10-year exclusive agreement to recycle e-VAC's manufacturing swarf, Cyclic ensures a steady, local flow of high-grade feedstock, while e-VAC eliminates the extreme costs of hazardous waste disposal.³⁹ This symbiotic relationship drastically reduces logistical overhead, lowers carbon emissions, and creates a highly resilient micro-economy that standardizes production costs.

Strategic Impediments and Future Outlook

While the unprecedented capitalization and rapid facility deployment of these post-2014 startups are historic, the sector faces profound strategic impediments that must be navigated perfectly over the coming decade to avoid collapse.

Price Dumping and State-Sponsored Market Manipulation

The primary existential threat to this nascent ecosystem is predatory pricing by established foreign monopolies. As seen historically in the rare earth sector, when non-Chinese ventures attempt to reach operational scale, state-backed entities can dramatically flood the market and lower global commodity prices, rendering western startups immediately economically unviable.⁵

To survive this inevitable pricing warfare, startups like Vulcan, Noveon, and USA Rare Earth require fiercely insulated, guaranteed demand structures. This dynamic explains why the DoD's 2027 DFARS procurement ban is the absolute linchpin of the entire industry. By completely outlawing Chinese magnets in lucrative defense platforms, the U.S. government has effectively created a captive, premium market where domestic startups can command higher pricing necessary to recoup their massive, billion-dollar capital expenditures without being undercut by subsidized foreign rivals.¹³

The Scaling Conundrum (Navigating the Valley of Death)

Transitioning from pilot-scale production (e.g., 10 to 100 metric tons) to massive commercial scale (10,000 metric tons) introduces severe engineering risks that have historically destroyed promising hardware startups, as evidenced by the failure of FeNix Magnetics.²⁶

Sintering NdFeB magnets or successfully aligning Iron Nitride requires highly precise, near-perfect thermal, chemical, and atmospheric controls. A failure to maintain exact tolerances at massive volumes results in compromised magnetic flux and unsellable product. Investors like Altimeter Capital, OneIM, and Ara Partners are betting heavily that companies like Vulcan, Noveon, and e-VAC can cross this chasm, but history shows an exceedingly high attrition rate in scaling deep-tech hardware.¹⁵

The Generational Human Capital Deficit

Because the United States actively offshored its magnet industry over two decades ago, there is currently a massive, generational gap in domestic metallurgical, chemical, and material science expertise.⁵ Startups are forced to recruit aggressively from adjacent, highly

competitive sectors (like advanced aerospace or semiconductor manufacturing) or rely entirely on specialized academic founders, as seen with Niron Magnetics' heavy reliance on the University of Minnesota's talent pipeline.¹⁹

The long-term commercial success of the sector will require robust, heavily funded public-private partnerships with research universities to continuously train the next generation of rare-earth metallurgists and magnetic engineers. Without a continuous influx of specialized talent, the physical infrastructure being built today will remain underutilized.

Conclusion

The United States is currently undertaking one of the most aggressive, highly capitalized, and strategically urgent industrial re-shoring initiatives in its history. Realizing the existential macroeconomic and military threat posed by an over-reliance on fragile, adversarial foreign supply chains for permanent magnets, a dynamic new sector of post-2014 startups has rapidly materialized to fill the void.

Crucially, these new companies are not simply attempting to mimic legacy Chinese manufacturing techniques. Instead, they are leveraging distinctly American advantages to leapfrog the status quo: advanced material science innovations that eliminate rare earths entirely (Niron Magnetics), proprietary circular-economy recycling technologies that tap into urban mines (Noveon and Cyclic Materials), and massive influxes of highly subsidized private capital merging with aggressive defense policy to build mega-factories (Vulcan Elements, e-VAC, USA Rare Earth, and MP Materials' 10X campus).

The successful commercialization and scaling of these highly ambitious startups over the next five years will fundamentally dictate the trajectory of the American defense industrial base, the speed of the global transition to electrified mobility, and the physical resilience of the artificial intelligence revolution. If these new entities can successfully navigate the "Valley of Death," overcome severe talent shortages, and withstand brutal external pricing pressures, they will have engineered a monumental shift in global power dynamics, permanently securing the supply chain of the 21st century's most critical hardware technologies.

Works cited

1. POSCO International Launches First US Rare Earth, Magnet Production Complex, accessed May 21, 2026, <https://en.sedaily.com/finance/2026/05/22/posco-international-launches-first-us-rare-earth-magnet>
2. U.S. Government Inks Deal with Rare-Earth Magnet Startups - NAM, accessed May 21, 2026, <https://nam.org/u-s-government-inks-deal-with-rare-earth-magnet-startups-35115/>
3. How a Minneapolis company could transform manufacturing of essential magnets for consumer electronics, clean energy transition - CBS News, accessed May 21, 2026, <https://www.cbsnews.com/minnesota/news/niron-magnetics-minneapolis-electronic-manufacturing/>

4. The History and Future of Rare Earth Magnets, accessed May 21, 2026, <https://noveon.co/historyandfuture>
5. US Magnet Business History, accessed May 21, 2026, <http://www.usmagneticmaterials.com/us-magnet-business-history.htm>
6. AML Superconductivity and Magnetics - Company, United States, accessed May 21, 2026, <https://www.swfinstitute.org/profile/5e39a66cfcbe7e8ca72a9c25>
7. U.S. Company Paves the Way for Permanent Magnets from Wide Array of Magnet Materials - PR Newswire, accessed May 21, 2026, <https://www.prnewswire.com/news-releases/us-company-paves-the-way-for-permanent-magnets-from-wide-array-of-magnet-materials-302425347.html>
8. BUILDING RESILIENT SUPPLY CHAINS, REVITALIZING AMERICAN MANUFACTURING, AND FOSTERING BROAD-BASED GROWTH - Biden White House, accessed May 21, 2026, https://bidenwhitehouse.archives.gov/wp-content/uploads/2021/06/100-day-supply-chain-review-report.pdf?utm_source=sfmc%E2%80%8B&utm_medium=email%E2%80%8B&utm_campaign=20210610_Global_Manufacturing_Economic_Update_June_Members
9. Vulcan Elements - Wikipedia, accessed May 21, 2026, https://en.wikipedia.org/wiki/Vulcan_Elements
10. University-born company Niron Magnetics receives \$17.5M DOE grant | College, accessed May 21, 2026, <https://cse.umn.edu/college/news/university-born-company-niron-magnetics-receives-175m-doe-grant>
11. Grant FA86502325513 E-Vac Magnetics - HigherGov, accessed May 21, 2026, <https://www.highergov.com/grant/FA86502325513/>
12. Department of Defense Enters an Agreement to Expand Domestic Manufacturing to Strengthen U.S. Supply Chains for Rare Earth Magnets, accessed May 21, 2026, <https://www.war.gov/News/Releases/Release/Article/3529874/departments-of-defense-enters-an-agreement-to-expand-domestic-manufacturing-to-s/>
13. Rare-Earth Magnet Maker Key For Pentagon Policy Wins New Funding - Vulcan Elements, accessed May 21, 2026, <https://vulcanelements.com/rare-earth-magnet-maker-key-for-pentagon-policy-wins-new-funding/>
14. About Us - USARE, accessed May 21, 2026, <https://www.usare.com/about/>
15. Vulcan Elements Announces \$65 Million Series A, and Produces High-Performance Rare Earth Magnets in the United States, accessed May 21, 2026, <https://vulcanelements.com/vulcan-elements-announces-65-million-series-a-and-produces-high-performance-rare-earth-magnets-in-the-united-states/>
16. Ara Partners Secures \$335 Million of Non-Recourse Financing for Permanent Magnet Manufacturing Facility, accessed May 21, 2026, <https://www.arapartners.com/news/ara-partners-secures-335-million-of-non-recourse-financing-for-permanent-magnet-manufacturing-facility/>
17. EcoFlux: Securing a Sustainable Future - Noveon Magnetics — Noveon Magnetics, accessed May 21, 2026, <https://noveon.co/forbeshighlight>

18. UMN Startup Niron Magnetics Featured in Forbes Magazine | Research & Innovation Office, accessed May 21, 2026,
<https://research.umn.edu/units/techcomm/news/umn-startup-niron-magnetics-featured-forbes-magazine>
19. Niron Magnetics, company that rose from Professor Jian-Ping Wang's research, awarded \$17.5M DOE grant - College of Science & Engineering - University of Minnesota Twin Cities, accessed May 21, 2026,
<https://cse.umn.edu/ece/news/niron-magnetics-company-rose-professor-jian-ping-wangs-research-awarded-175m-doe-grant>
20. Professor Jian-Ping Wang: imagining a sustainable future through Niron Magnetics - College of Science & Engineering - University of Minnesota Twin Cities, accessed May 21, 2026,
<https://cse.umn.edu/ece/news/professor-jian-ping-wang-imagining-sustainable-future-through-niron-magnetics>
21. Niron Magnetics: Home, accessed May 21, 2026, <https://nironmagnetics.com/>
22. Commercializing the Clean Earth Magnet technology - Saint Mary's University of Minnesota, accessed May 21, 2026,
<https://www.smumn.edu/newsroom/commercializing-the-clean-earth-magnet-technology/>
23. Niron Magnetics and Stellantis Partner to Pioneer Rare-Earth-Free Electric Motor Innovation, accessed May 21, 2026,
<https://nironmagnetics.com/niron-magnetics-and-stellantis-partner-to-pioneer-rare-earth-free-electric-motor-innovation/>
24. Advanced Manufacturing of Alpha Double Prime Iron Nitride (ADPIN): An Innovative Rare Earth Element (REE) Free Ultra-High Performer, accessed May 21, 2026,
<https://www.energy.gov/documents/O50-projects19-advanced-manufacturing-alpha-double-prime-iron-nitride-fenix-magneticspdf>
25. University of Minnesota Startup Breaks Ground on First Full-Scale Magnet Facility in Minnesota, accessed May 21, 2026,
<https://research.umn.edu/units/techcomm/news/university-minnesota-startup-breaks-ground-first-full-scale-magnet-facility-minnesota>
26. Fenix Magnetics 2026 Company Profile: Valuation, Funding & Investors | PitchBook, accessed May 21, 2026,
<https://pitchbook.com/profiles/company/167035-60>
27. Final Report - OSTI, accessed May 21, 2026,
<https://www.osti.gov/servlets/purl/1866484>
28. ITO Funded Projects | Department of Energy, accessed May 21, 2026,
<https://www.energy.gov/cmei/ito/ito-funded-projects>
29. AMMTO Funded Projects | Department of Energy, accessed May 21, 2026,
<https://www.energy.gov/cmei/ammto/ammto-funded-projects>
30. Ohio Federal Research Network Round 1 Centers of Excellence solicitation, accessed May 21, 2026,
<https://ohiofrn.org/solicitations/ohio-federal-research-network-round-1-centers-excellence-solicitation>

31. Noveon Magnetics Raises \$215 Million In Series C Funding Round ..., accessed May 21, 2026, <https://www.superbcrew.com/noveon-magnetics-raises-215-million-in-series-c-funding-round/>
32. About - Noveon Magnetics, accessed May 21, 2026, <https://noveon.co/about>
33. EcoFlux: Securing a Sustainable Future - Noveon Magnetics, accessed May 21, 2026, <https://noveon.co/abj2>
34. EcoFlux: Securing a Sustainable Future - Noveon Magnetics — Noveon Magnetics, accessed May 21, 2026, <https://noveon.co/san-marcos-company-aims-to-energize-us-magnets-supply>
35. Texas Startup Targets Rare Earth Magnet Production, accessed May 21, 2026, <https://www.apexmagnets.com/news-how-tos/texas-based-company-targets-rare-earth-magnet-production-apex-magnets/>
36. Noveon Magnetics, accessed May 21, 2026, <https://noveon.co/>
37. Cyclic Materials and VACUUMSCHMELZE Expand Partnership to Recycle Rare Earth Magnet Manufacturing Waste in the U.S., accessed May 21, 2026, <https://cyclicmaterials.earth/resources/cyclic-materials-and-vacuumschmelze-expand-partnership-to-recycle-rare-earth-magnet-manufacturing-waste-in-the-u-s>
38. Cyclic Materials Announces USD \$25M Investment to Establish Centre of Excellence for Rare Earth Recycling in Kingston, Ontario, accessed May 21, 2026, <https://cyclicmaterials.earth/resources/cyclic-materials-announces-usd-25m-investment-to-establish-centre-of-excellence-for-rare-earth-recycling-in-kingston-ontario>
39. Cyclic Materials to Expand U.S. Rare Earth Recycling Footprint with New South Carolina Campus, accessed May 21, 2026, <https://cyclicmaterials.earth/resources/cyclic-materials-to-expand-u-s-rare-earth-recycling-footprint-with-new-south-carolina-campus>
40. Cyclic Materials and VAC Partner to Recycle Magnet Manufacturing By-products, accessed May 21, 2026, <https://cyclicmaterials.earth/resources/cyclic-materials-and-vac-partner-to-recycle-magnet-manufacturing-by-products>
41. USA Rare Earth commissions new magnet production line at Oklahoma plant - MINING.COM, accessed May 21, 2026, <https://www.mining.com/usa-rare-earth-commissions-new-magnet-production-line-at-oklahoma-plant/>
42. Vulcan Elements Inc. - | SBIR, accessed May 21, 2026, <https://www.sbir.gov/portfolio/2451639>
43. Vulcan Elements Holds Grand Opening of Small-Scale Commercial Manufacturing and R&D Facility, accessed May 21, 2026, <https://vulcanelements.com/vulcan-elements-holds-grand-opening-of-small-scale-commercial-manufacturing-and-rd-facility/>
44. US rare earth magnet startup raises \$65M to scale up production - MINING.COM, accessed May 21, 2026, <https://www.mining.com/us-rare-earth-magnet-startup-raises-65m-to-scale-up->

[production/](#)

45. ReElement part of \$1.4B U.S. Defense partnership to scale domestic rare earth magnet production - YouTube, accessed May 21, 2026, <https://www.youtube.com/watch?v=KtD5Pgn3378>
46. BENSON WILL BE HOME TO \$1 BILLION RARE EARTH MAGNET FACILITY, accessed May 21, 2026, <https://www.growwithjoco.com/news/benson-will-be-home-to-1-billion-rare-earth-magnet-facility/>
47. Vulcan Elements to open \$918 million plant with plans for 1000 jobs in Johnston County, accessed May 21, 2026, <https://www.wral.com/news/local/vulcan-elements-918-million-plant-1000-jobs-johnston-county-november-2025/>
48. Vulcan Elements Selects Johnston County for 1000-Job Magnet Factory, accessed May 21, 2026, <https://researchtriangle.org/news/vulcan-elements-selects-johnston-county-for-1000-job-magnet-factory/>
49. USA Rare Earth - Wikipedia, accessed May 21, 2026, https://en.wikipedia.org/wiki/USA_Rare_Earth
50. Pini Althaus – MINEX Eurasia 2024, accessed May 21, 2026, <https://2024.minexeurasia.com/forum-speakers/pini-althaus/>
51. Rare Earth Magnet Producer To Start Operations In Stillwater - Site Selection Magazine, accessed May 21, 2026, <https://siteselection.com/rare-earth-magnet-producer-to-start-operations-in-stillwater/>
52. USA Rare Earth, a Domestic Rare Earth Mining & Magnet Company, to Become a Publicly Traded Company via Business Combination with Inflection Point Acquisition Corp. II, accessed May 21, 2026, <https://inflectionpointacquisition.com/IPAX-Investor-Relations/press-release-details/2024/USA-Rare-Earth-a-Domestic-Rare-Earth-Mining--Magnet-Company-to-Become-a-Publicly-Traded-Company-via-Business-Combination-with-Inflection-Point-Acquisition-Corp.-II/default.aspx>
53. Deal rush in rare earths as west seeks to loosen China's grip, accessed May 21, 2026, <https://www.ft.com/content/0c05ef4c-55eb-4293-8bba-7985e4c25e78?syn-25a6b1a6=1>
54. USA Rare Earth wins up to \$19.3M in DOE backing - MINING.COM, accessed May 21, 2026, <https://www.mining.com/usa-rare-earth-wins-up-to-19-3m-in-doe-backing/>
55. e-VAC Magnetism Receives Historic \$111.9 Million Qualifying Advanced Energy Project Tax Credit for Construction of First US Manufacturing Facility in South Carolina - Vacuumschmelze, accessed May 21, 2026, <https://vacuumschmelze.com/Newsroom/e-vac-magnetism-receives-historic-1119-million-qualifying-advanced-energy-project-tax-credit-for-construction-of-first-us-manufacturing-facility-in-south-carolina-n2641>
56. eVAC Magnetism, accessed May 21, 2026, <https://e-vac-magnetism.com/>

57. eVAC - Advanced Magnetic Solutions | VAC, accessed May 21, 2026,
<https://evac-magnetics.com/>
58. eVAC Wins Energy Transition Deal of the Year for U.S. Permanent ..., accessed May 21, 2026,
<https://www.arapartners.com/news/evac-wins-energy-transition-deal-of-the-year-for-u-s-permanent-magnet-manufacturing-facility/>
59. eVAC Wins Energy Transition Deal of the Year for U.S. Permanent Magnet Manufacturing Facility - PR Newswire, accessed May 21, 2026,
<https://www.prnewswire.com/news-releases/evac-wins-energy-transition-deal-of-the-year-for-us-permanent-magnet-manufacturing-facility-302404577.html>
60. MP Materials Metal & Magnet Production Facility - Fluor, accessed May 21, 2026,
<https://www.fluor.com/projects/mp-materials-metal-magnet-production-facility>
61. Independence | MP Materials, accessed May 21, 2026,
<https://mpmaterials.com/independence>
62. MP Materials To Build '10X,' a \$1.25B Rare Earth Magnet Manufacturing Campus in North Texas - AllianceTexas Mobility Innovation Zone, accessed May 21, 2026,
<https://www.alliancetexasbiz.com/news/mp-materials-to-build-10x-a-1-25b-rare-earth-magnet-manufacturing-campus-in-north-texas/>
63. MP Materials to build \$1.25 billion rare earth magnet factory near Fort Worth | Hillwood, a Perot Company., accessed May 21, 2026,
<https://www.hillwood.com/newsroom/news-articles/mp-materials-to-build-1-25-billion-rare-earth-magnet-factory-near-fort-worth/>
64. Governor Abbott Announces MP Materials Rare Earth Magnet Manufacturing Expansion In Northlake, accessed May 21, 2026,
<https://gov.texas.gov/news/post/governor-abbott-announces-mp-materials-rare-earth-magnet-manufacturing-expansion-in-northlake>
65. MP Materials Selects Northlake, Texas, as the Site of "10X," a New U.S. Rare Earth Magnet Manufacturing Campus, accessed May 21, 2026,
<https://mpmaterials.com/news/mp-materials-selects-northlake-texas-as-the-site-of-10x-a-new-us-rare-earth-magnet-manufacturing-campus/>
66. Company - ReElement Technologies, accessed May 21, 2026,
<https://www.reelementtech.com/v3/company>
67. ReElement Technologies, accessed May 21, 2026,
<https://www.reelementtech.com/>
68. POSCO International bets big on US rare earth supply chain, accessed May 21, 2026,
<https://www.koreatimes.co.kr/business/companies/20260522/posco-international-bets-big-on-us-rare-earth-supply-chain>
69. POSCO International to Build First Integrated Rare Earth and Permanent Magnet Production Complex in the U.S., accessed May 21, 2026,
<https://www.asiae.co.kr/en/article/distribution/2026052207493773083>
70. Posco International, ReElement build US rare earth-to-magnet complex to bolster Korea-US supply chain - CHOSUNBIZ, accessed May 21, 2026,
<https://biz.chosun.com/en/en-industry/2026/05/22/HBAJCQSNEBCIZNANH2ZW3I2OFQ/>

71. Fluxworks closes \$5M funding to commercialize Celestial Gear - InnovationMap, accessed May 21, 2026,
<https://houston.innovationmap.com/fluxworks-celestial-gear-seed-round-2675040089.html>